A dimension-independent spectral rounding bound for orthogonal synchronization*

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Abstract

Spectral methods for orthogonal synchronization relax the orthogonality constraints and then round the resulting matrix blocks to the orthogonal group. We bound the objective value after this rounding directly in terms of the spectral relaxation. On any finite connected weighted graph with labels in $\mathrm{O}(r)$, let γ be the normalized scalar Laplacian gap and let Λ_r be the sum of the lowest r normalized connection Laplacian eigenvalues. Nearest-orthogonal rounding of the degree-normalized bottom spectral frame gives normalized squared-Frobenius frustration $\nu(Q) \leq (3 + 12/\gamma)\Lambda_r/r$. The coefficient is independent of r and depends on the graph only through γ. The proof controls the full positive magnitudes of the spectral blocks through a classical matrix absolute-value inequality, graph Poincaré and the frame's second-moment normalization. We compare this certificate with the dimension-dependent connection-Cheeger bound and with a dimension-independent deduction from Frobenius eigenspace perturbation estimates. The latter has additional graph dependence and bounds the objective in terms of the unknown optimum. No closeness to a consistent assignment is assumed.

Keywords. Orthogonal synchronization; connection Laplacian; spectral rounding; graph spectral gap; polar decomposition; approximation guarantees.

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1 Introduction

Orthogonal synchronization assigns an orthogonal matrix to each vertex of a graph so that neighboring assignments agree, as closely as possible, with prescribed relative transformations. A spectral relaxation replaces the orthogonality constraint at every vertex by one global normalization. Its lowest eigenvectors minimize the relaxed energy, but their individual blocks need not be orthogonal. The approximation guarantee depends on how much energy is added when those blocks are projected back to the group.

Bandeira, Singer, and Spielman related this rounding problem to the normalized connection spectrum and the ordinary graph gap [1]. Let r be the matrix dimension, let γ be the gap of the normalized scalar Laplacian, and let Λ_r be the sum of the lowest r normalized connection eigenvalues. In the normalization used below, their Theorem 2.6 bounds the minimum frustration ν_* by

$$\frac{\Lambda_r}{r} \leq \nu_* \leq \frac{1026r^4}{\gamma}\frac{\Lambda_r}{r}. \tag{1.1}$$

The lower quantity is the value of the spectral relaxation. The upper coefficient grows with r, even when the graph gap stays fixed.

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Dimension-independent guarantees can also be obtained from eigenspace perturbation estimates. Such estimates underlie the synchronization analyses of Liu, Yue, and So [2] and Doherty, Rosen, and Leonard [3]. Applying the Frobenius bound of Yu, Wang, and Samworth [4] to a consistent reference built from an optimizer gives a comparison for the same normalized rounding construction considered here. The resulting coefficient is independent of r, but has a squared inverse-gap dependence and an additional graph factor. On an unweighted d-regular graph with n vertices, this factor is n/d. Section 5 derives this consequence explicitly; it is not a theorem quoted from those synchronization papers.

We prove that nearest-orthogonal rounding of the normalized bottom spectral frame produces a field Q satisfying

$$\nu(Q) \le \left(3 + \frac{12}{\gamma}\right) \frac{\Lambda_r}{r}. \tag{1.2}$$

Thus the dimension factor in (1.1) is removed, while the graph dependence remains through γ alone. Unlike the optimum-relative perturbation comparison just described, (1.2) has the observed relaxation value on its right-hand side. It therefore bounds the loss from the relaxed optimum to the rounded solution using quantities supplied by the spectral construction. In particular, on bounded-degree graph families with a uniformly positive normalized gap, the approximation factor remains uniform in both the number of vertices and the matrix dimension. The guarantee measures agreement with the supplied edge transformations; no model for a latent true assignment is required.

The proof keeps the full positive magnitude $|X_i| = (X_i^T X_i)^{1/2}$ of each spectral block. Its distance from I_r is exactly the distance of X_i from the orthogonal group. Orthogonal transport on the left disappears from this magnitude, so the classical Hilbert–Schmidt inequality of Araki and Yamagami [5] and graph Poincaré control its variation across the vertices. The global identity $\mathbb{E}_\mu |X_i|^2 = I_r$ then controls the distance of its mean from I_r. This avoids estimating individual column norms and pairwise inner products separately, as in the dimension-dependent argument of [1]. The matrix inequality itself is established and has already been used in synchronization by Zhu, Wang, and So [6]; Lemma 3.2 gives its global consequence for rounding an arbitrary normalized matrix field.

2 The spectral construction and its guarantee

Let G be a finite connected undirected graph on $n \ge 2$ vertices, with weights $w_{ij} = w_{ji} \ge 0$ and $w_{ii} = 0$. Connectivity refers to the positive-weight edges. Define

$$a_i = \sum_j w_{ij} > 0, \qquad v = \sum_i a_i, \qquad \mu_i = \frac{a_i}{v}, \qquad D = \operatorname{diag}(a_1, \dots, a_n).$$

For a matrix field Y, write $\mathbb{E}_\mu Y_i = \sum_i \mu_i Y_i$. Let $r \ge 1$ be an integer. On every positive-weight edge choose $\rho_{ij} \in \mathrm{O}(r)$ with $\rho_{ji} = \rho_{ij}^T$. Writing $W = (w_{ij})$, $W_\rho = (w_{ij}\rho_{ij})_{i,j}$, and $D_\rho = D \otimes I_r$, set

$$\mathcal{L}_0 = I_n - D^{-1/2} W D^{-1/2}, \qquad \mathcal{L}_\rho = I_{nr} - D_\rho^{-1/2} W_\rho D_\rho^{-1/2}.$$

Eigenvalues are ordered increasingly and counted with multiplicity. Put

$$\gamma = \lambda_2(\mathcal{L}_0) > 0, \qquad \Lambda_r = \sum_{k=1}^{r} \lambda_k(\mathcal{L}_\rho).$$

The scalar gap γ measures how the underlying weighted graph controls variation of a vertex function. The connection eigenvalues incorporate the edge transformations and determine the relaxed objective.

For a field of square matrices $X = (X_i)_{i=1}^n$, use the energy

$$\mathcal{E}_\rho(X) = \frac{1}{2v} \sum_{i,j} w_{ij} \, \|X_i - \rho_{ij} X_j\|_F^2 \,, \tag{2.1}$$

where the sum includes both orientations of every edge. For an orthogonal field $Q \in \mathrm{O}(r)^n$, define

$$\nu(Q) = \frac{\mathcal{E}_\rho(Q)}{r}, \qquad \nu_* = \min_{Q \in \mathrm{O}(r)^n} \nu(Q).$$

The minimum exists because the domain is compact.

Choose orthonormal eigenvectors $z^1, \dots, z^r \in \mathbb{R}^{nr}$ for the lowest r eigenvalues of $\mathcal{L}_\rho$. If $z_i^k \in \mathbb{R}^r$ is the block at vertex i, form

$$X_i = \sqrt{\frac{v}{a_i}} \, [\, z_i^1 \ \cdots \ z_i^r \,]. \tag{2.2}$$

This scaling gives $\mathbb{E}_\mu X_i^T X_i = I_r$. For any full singular value decomposition $X_i = U_i \Sigma_i V_i^T$, with $U_i, V_i \in \mathrm{O}(r)$, choose

$$Q_i = U_i V_i^T. \tag{2.3}$$

This is a nearest orthogonal matrix to X_i in Frobenius norm, with a full orthogonal completion even when X_i is singular. It agrees with the usual polar rule on invertible blocks. The singular convention differs from the identity fallback allowed in Algorithm 2.5 of [1]; Remark 4.1 explains its role in the proof.

Theorem 2.1. *For every graph and edge assignment satisfying the assumptions above, every choice of the spectral frame in* (2.2)*, and every full-SVD rounding in* (2.3)*,*

$$\boxed{\frac{\Lambda_r}{r} \le \nu_* \le \nu(Q) \le \left(3 + \frac{12}{\gamma}\right) \frac{\Lambda_r}{r} \le \frac{18 \Lambda_r}{r\gamma}.} \tag{2.4}$$

In particular, $\nu(Q) \le (18/\gamma)\nu_*$*. If* $\nu_* = 0$*, then* $\nu(Q) = 0$*.*

The spectral frame need not be unique. The assertion holds for every orthonormal choice, including when $\lambda_r(\mathcal{L}_\rho) = \lambda_{r+1}(\mathcal{L}_\rho)$. All dependence on the graph in the approximation factor is through γ; there is no assumption that the labels are close to a consistent assignment.

The theorem gives both a spectral certificate and an approximation ratio. The first compares the rounded objective with the computable lower bound Λ_r/r; the second follows because this lower bound is no larger than ν_*. To see when the certificate is informative in absolute terms, note that every orthogonal field satisfies $\nu(Q) \le 2$, since each squared edge residual is at most $4r$. The upper bound in (2.4) improves this ceiling whenever

$$\frac{\Lambda_r}{r} < \frac{2\gamma}{3\gamma + 12}. \tag{2.5}$$

A small gap weakens the certificate, although no noise restriction is needed for its validity.

For an unweighted d-regular graph with $\gamma \ge \gamma_0 > 0$, the optimum-relative consequence is

$$\nu(Q) \le (3 + 12/\gamma_0)\nu_*.$$

It is uniform in n and r, including along a family with fixed degree and increasing size. Section 5 derives the corresponding Frobenius perturbation bound and compares the two estimates on regular graph families.

3 Control of the matrix magnitudes

To control the change in objective after rounding, we first need to control how far the relaxed blocks lie from the orthogonal group. Their positive magnitudes isolate this distance: if Q_i is a nearest orthogonal matrix to X_i, then $\|X_i - Q_i\|_F = \||X_i| - I_r\|_F$. A bound on the variation of $|X_i|$ alone would only place these magnitudes near a common matrix. The normalization $\mathbb{E}_\mu |X_i|^2 = I_r$ supplies the additional control that places that common matrix near I_r.

We carry this out with two estimates. The first passes from the connection energy to ordinary variation of the magnitudes; the second combines graph Poincaré with the normalization to bound the rounding error.

Lemma 3.1 (Hilbert–Schmidt absolute-value inequality)**.** *For real square matrices A, B of the same size,*

$$\||A| - |B|\|_F^2 \le 2\,\|A - B\|_F^2\,, \qquad |A| = (A^T A)^{1/2}. \tag{3.1}$$

Proof. For real symmetric matrices H, K, let (u_a, h_a) and (v_b, k_b) be orthonormal eigenbases and their eigenvalues. Expansion in these bases gives

$$\begin{aligned}\||H| - |K|\|_F^2 &= \sum_{a,b} (|h_a| - |k_b|)^2 |\langle u_a, v_b\rangle|^2 \\ &\le \sum_{a,b} (h_a - k_b)^2 |\langle u_a, v_b\rangle|^2 = \|H - K\|_F^2\,.\end{aligned}$$

Apply this inequality to the symmetric dilations

$$\mathcal{D}(A) = \begin{pmatrix} 0 & A \\ A^T & 0 \end{pmatrix}, \qquad |\mathcal{D}(A)| = \operatorname{diag}(|A^T|, |A|).$$

Since $\|\mathcal{D}(A) - \mathcal{D}(B)\|_F^2 = 2\,\|A - B\|_F^2$, it follows that

$$\||A| - |B|\|_F^2 + \left\||A^T| - |B^T|\right\|_F^2 \le 2\,\|A - B\|_F^2\,.$$

Discarding the second nonnegative term proves (3.1). This is a finite-dimensional verification of the inequality in [5]. □

The graph Poincaré inequality, applied entry by entry, states that every real matrix field P satisfies

$$\mathbb{E}_\mu\,\|P_i - \mathbb{E}_\mu P_i\|_F^2 \le \frac{1}{2v\gamma} \sum_{i,j} w_{ij}\,\|P_i - P_j\|_F^2\,. \tag{3.2}$$

Indeed, for each entry the centered field, multiplied at vertex i by $\sqrt{\mu_i}$, is orthogonal to the kernel vector $(\sqrt{\mu_i})_{i=1}^n$ of $\mathcal{L}_0$; its Rayleigh quotient is at least γ.

Lemma 3.2. *Suppose $\mathbb{E}_\mu X_i^T X_i = I_r$. For each i, let Q_i be any full-SVD rounding of X_i. Then*

$$\mathbb{E}_\mu\,\|X_i - Q_i\|_F^2 \le \frac{4}{\gamma}\mathcal{E}_\rho(X).$$

Proof. Set $P_i = |X_i|$ and $M = \mathbb{E}_\mu P_i$. Since left multiplication by an orthogonal matrix preserves the positive magnitude, $|\rho_{ij} X_j| = P_j$. Lemma 3.1 gives

$$\frac{1}{2v} \sum_{i,j} w_{ij}\,\|P_i - P_j\|_F^2 \le 2\mathcal{E}_\rho(X).$$

By (3.2),

$$V_P := \mathbb{E}_\mu\,\|P_i - M\|_F^2 \le \frac{2}{\gamma}\mathcal{E}_\rho(X).$$

It remains to control the mean M, since small variation around M does not by itself control distance from I_r. The assumption on X says $\mathbb{E}_\mu P_i^2 = I_r$. Expanding the matrix square and using $\mathbb{E}_\mu P_i = M$ yields

$$\mathbb{E}_\mu (P_i - M)^2 = I_r - M^2 \succeq 0.$$

No commutativity between different P_i is needed in this identity. Because $M \succeq 0$, its eigenvalues lie in $[0, 1]$. Consequently,

$$\|I_r - M\|_F^2 \leq \operatorname{tr}(I_r - M^2) = V_P.$$

The Hilbert-space variance identity now gives

$$\mathbb{E}_\mu \|P_i - I_r\|_F^2 = V_P + \|M - I_r\|_F^2 \leq 2V_P \leq \frac{4}{\gamma}\mathcal{E}_\rho(X).$$

Finally, if $X_i = U_i \Sigma_i V_i^T$ and $Q_i = U_i V_i^T$, then

$$\|X_i - Q_i\|_F^2 = \|\Sigma_i - I_r\|_F^2 = \|P_i - I_r\|_F^2 .$$

This equality also holds for singular blocks and completes the proof. □

4 Proof of the spectral rounding bound

The orthonormality of the eigenvectors in (2.2) and the quadratic form of $\mathcal{L}_\rho$ give

$$\mathbb{E}_\mu X_i^T X_i = I_r, \qquad \mathcal{E}_\rho(X) = \Lambda_r.$$

Thus Lemma 3.2 bounds the mean squared rounding error by $4\Lambda_r/\gamma$. To transfer this bound to the edge energy, decompose each residual as

$$Q_i - \rho_{ij} Q_j = (Q_i - X_i) + (X_i - \rho_{ij} X_j) + \rho_{ij}(X_j - Q_j).$$

The squared norm of a sum of three vectors is at most three times the sum of their squared norms. Summing with the weights in (2.1), and using the symmetry of w_{ij}, gives

$$\begin{aligned}\mathcal{E}_\rho(Q) &\leq 3\mathcal{E}_\rho(X) + 3\mathbb{E}_\mu \|X_i - Q_i\|_F^2 \\ &\leq \left(3 + \frac{12}{\gamma}\right)\Lambda_r.\end{aligned}$$

Division by r proves the first upper bound in (2.4). Since the normalized scalar Laplacian has spectrum in $[0, 2]$, we have $\gamma \leq 2$ and $3 + 12/\gamma \leq 18/\gamma$.

For the lower bound, take any orthogonal field Q and form the nr-by-r matrix Z with block rows $\sqrt{\mu_i} Q_i$. Then $Z^T Z = I_r$, while

$$\operatorname{tr}(Z^T \mathcal{L}_\rho Z) = \mathcal{E}_\rho(Q).$$

The minimum-trace variational principle implies $\Lambda_r \leq \mathcal{E}_\rho(Q)$. Minimize over Q and divide by r. Combining this lower bound with the upper bound gives the approximation ratio and the zero-optimum assertion. □

Remark 4.1 (Singular blocks and the scope of rounding). The nearest-orthogonal convention is necessary for the displayed rounding-error identity. For $X = \operatorname{diag}(-1, \ldots, -1, 0)$, the squared distance to $\mathrm{O}(r)$ is 1, whereas $\|X - I_r\|_F^2 = 4(r-1) + 1$. This explains why the proof cannot replace a singular block arbitrarily by I_r. It does not assert that the identity convention violates the bound proved in [1]. Theorem 2.1 concerns $\mathrm{O}(r)$; no determinant correction for $SO(r)$ or projection to another subgroup is included.

5 Comparison with perturbation bounds

Theorem 2.1 bounds the rounded objective by the observed spectral relaxation value. To compare it with reference-based estimates, we apply a standard eigenspace perturbation bound to the same normalized construction. The calculation shows both what that approach already gives and which additional graph dependence the magnitude argument avoids.

5.1 A normalized perturbation comparison on every graph

Let $H \in \mathrm{O}(r)^n$ minimize the objective, and write

$$F(Q) = \sum_{i<j} w_{ij} \left\| \rho_{ij} - Q_i Q_j^T \right\|_F^2 = vr\nu(Q), \qquad F_* = F(H) = vr\nu_*.$$

Converting the label error to the Frobenius norm of the normalized Laplacian introduces the graph factor

$$\kappa(G) = v \max_{w_{ij}>0} \frac{w_{ij}}{a_i a_j}. \tag{5.1}$$

It is unchanged by a common rescaling of the weights. For an unweighted d-regular graph, $a_i = d$ and $v = nd$, so $\kappa(G) = n/d$.

Replace the measured labels by the consistent labels $H_i H_j^T$ and let $\mathcal{L}_H$ be the resulting normalized connection Laplacian. With $\mathsf{H} = \mathrm{diag}(H_1, \dots, H_n)$, one has

$$\mathcal{L}_H = \mathsf{H}(\mathcal{L}_0 \otimes I_r)\mathsf{H}^T.$$

Its kernel therefore has the orthonormal frame Z_H with block rows $\sqrt{\mu_i} H_i$, and its first positive eigenvalue is γ. For $\Delta = \mathcal{L}_\rho - \mathcal{L}_H$, direct expansion gives

$$\begin{aligned} \|\Delta\|_F^2 &= 2 \sum_{i<j} \frac{w_{ij}^2}{a_i a_j} \left\| \rho_{ij} - H_i H_j^T \right\|_F^2 \\ &\le 2 \frac{\kappa(G)}{v} F_* = 2\kappa(G) r \nu_*. \end{aligned} \tag{5.2}$$

Let Z be the observed bottom spectral frame, whose block rows are $\sqrt{\mu_i} X_i$. Theorem 2 of [4], applied to $-\mathcal{L}_H$ and $-\mathcal{L}_\rho$, provides an $S \in \mathrm{O}(r)$ such that

$$\|Z - Z_H S\|_F \le \frac{2\sqrt{2}}{\gamma} \|\Delta\|_F . \tag{5.3}$$

Only the gap of the reference matrix is required. The Frobenius branch of this estimate also appears in Theorem 5 of [2] and Theorem 8 of the extended manuscript of [3]. Since $H_i S$ is orthogonal and Q_i is a nearest orthogonal matrix to X_i, the triangle inequality gives $\|Q_i - H_i S\|_F \le 2 \|X_i - H_i S\|_F$. Thus the degree factors in the frame and the vertex weights agree, and

$$R := \sum_i \mu_i \|Q_i - H_i S\|_F^2 \le \frac{32}{\gamma^2} \|\Delta\|_F^2 \le \frac{64\kappa(G) r}{\gamma^2} \nu_*. \tag{5.4}$$

To return from distance to the reference field to objective value, put $T_i = H_i^T Q_i S^T$. Then $\left\| Q_i Q_j^T - H_i H_j^T \right\|_F = \|T_i - T_j\|_F$ and $R = \sum_i \mu_i \|T_i - I_r\|_F^2$. Hence

$$F(Q) \le 2F_* + 2 \sum_{i<j} w_{ij} \|T_i - T_j\|_F^2 ,$$

$$\sum_{i<j} w_{ij} \|T_i - T_j\|_F^2 \le 2 \sum_i a_i \|T_i - I_r\|_F^2 = 2vR.$$

After division by vr, (5.4) yields the comparison

$$\nu(Q) \le \left(2 + \frac{256\kappa(G)}{\gamma^2}\right)\nu_*. \tag{5.5}$$

This is a consequence of the stated Frobenius perturbation estimate, not a new perturbation theorem. It applies to the same normalized $\mathrm{O}(r)$ rounding as Theorem 2.1, including singular blocks, and requires no small-perturbation assumption.

The coefficient in (5.5) is independent of r. The differences from (2.4) are its additional parameter $\kappa(G)$, its squared inverse-gap dependence, and its right-hand side involving ν_*. The lower bound $\Lambda_r/r \le \nu_*$ cannot replace ν_* by Λ_r/r in this upper estimate. The comparison concerns the direct Frobenius deduction above; it does not exclude sharper perturbation arguments under other estimates or additional assumptions.

5.2 Two graph families

For fixed-degree regular graph families with $\gamma \ge \gamma_0 > 0$, (5.5) has a coefficient that grows with n, while (2.4) gives a uniform factor. A separate noncomplete example shows when the spectral certificate improves the universal ceiling. For $K_{m,m}$ with $m \ge 2$, the normalized scalar spectrum is $0, 1, \ldots, 1, 2$, so $\gamma = 1$ and $\kappa(G) = 2$. Theorem 2.1 and (5.5) give, respectively,

$$\nu(Q) \le 15\frac{\Lambda_r}{r}, \qquad \nu(Q) \le 514\nu_*.$$

The first bound is below 2 when $\Lambda_r/r < 2/15$. By comparison, the displayed BSS upper bound in (1.1) is already at least 2 when $\Lambda_r/r \ge 2/(1026r^4)$. These thresholds compare the stated guarantees; they do not assert that every intermediate spectral value occurs or that the actual rounded solutions have large error.

One can also express the guarantee in terms of corrupted edge weight. Suppose labels agree with a consistent orthogonal field except on edges carrying an ε fraction of the total unordered edge weight, with arbitrary orthogonal labels on those edges. Testing that field gives $\nu_* \le 2\varepsilon$, since each affected squared residual is at most $4r$. On $K_{m,m}$, Theorem 2.1 consequently gives $\nu(Q) \le 30\varepsilon$, uniformly in r. This controls agreement with the supplied labels; it is not an estimate of distance to the field before corruption.

5.3 The complete-graph overlap

A dimension-independent complete-graph approximation follows more directly from the Frobenius branch of Theorem 5 of [2]. On the unweighted complete graph, use F and F_* as above. Write H and Q for the nr-by-r matrices obtained by stacking the blocks of the optimizer and the rounded field, respectively. Let C have off-diagonal blocks ρ_{ij} and diagonal blocks I_r. For $\Delta_C = C - HH^T$, one has $\|\Delta_C\|_F^2 = 2F_*$, while HH^T has eigengap n. The rounded spectral blocks are those of $\sqrt{n}\,V$, where V is a top-r orthonormal frame of C. This is the construction of Section 2 because $\mathcal{L}_\rho = I_{nr} - (C - I_{nr})/(n-1)$. The cited estimate followed by nearest projection gives

$$\min_{S \in \mathrm{O}(r)} \|Q - HS\|_F^2 \le \frac{32}{n}\,\|\Delta_C\|_F^2 = \frac{64F_*}{n}. \tag{5.6}$$

Choose S attaining the minimum in (5.6) and set $T_i = H_i^T Q_i S^T$. The complete-graph variance identity gives

$$F(Q) \le 2F_* + 2\sum_{i<j}\|T_i - T_j\|_F^2 \le 2F_* + 2n\sum_i \|T_i - I_r\|_F^2 \le 130F_*.$$

Thus dimension independence on this graph is already a consequence of the earlier estimate. Theorem 2.1 retains an upper bound in terms of the observed spectral relaxation on every connected weighted graph, with all graph dependence in its coefficient through γ.

Declarations

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